

Toy VELO Track Characterisation Study

Derived from `characterisation.ipynb`

Quantum Track Reconstruction

April 2, 2026

Abstract

This document summarises the workflow and analysis in `Toy_Characterisation/characterisation.ipynb`. The study evaluates segment-pair angle behaviour, track reconstruction quality, and scaling trends for a toy VELO model under varying event sizes, scattering strengths, and angular phase-space settings. The key tuning parameter is the segment-pair acceptance threshold ε , computed from detector and physics uncertainties.

1 Objectives

The notebook performs five main tasks:

1. Build reusable event-generation and angle-analysis helpers.
2. Solve reconstruction on a clean event and validate truth-reco matching.
3. Characterise pairwise segment-angle distributions on 10-event and 100×50-event ensembles.
4. Run a parametric scan versus event size for multiple scattering and generation-angle settings.
5. Produce acceptance histograms comparing true and false segment pairs.

2 Setup and Definitions

2.1 Geometry and Event Model

A planar detector geometry is generated with evenly spaced modules:

- Number of modules: configurable (clean-event baseline uses 5).
- First module position: $z_0 = 100$ mm.
- Module spacing: $\Delta z = 33$ mm.
- Half-widths: $l_x = l_y = 50$ mm.

Events are generated with primary vertices and charged pion tracks, with configurable angular bounds $\phi_{\max}$ and $\theta_{\max}$.

2.2 Hamiltonian Acceptance Scale

The acceptance threshold is computed as

$$\varepsilon = \sqrt{2\theta_s^2 + 12\theta_r^2 + 2\theta_{\min}^2}, \quad (1)$$

with

$$\theta_s = s \sigma_{\text{scatt}}, \quad \theta_r = \arctan\left(\frac{s \sigma_{\text{res}}}{\Delta z}\right), \quad (2)$$

where s is a sigma scale factor (default $s = 3$ in this notebook).

For the clean-event baseline:

- $\sigma_{\text{res}} = 0$,
- $\sigma_{\text{scatt}} = 10^{-4}$ rad,
- $\Delta z = 33$ mm,
- $s = 3$,

which gives approximately

$$\varepsilon \approx 4.24 \times 10^{-4} \text{ rad} \approx 0.424 \text{ mrad}. \quad (3)$$

2.3 Segment-Pair Angle Definition

The measured angle is the angle between consecutive segments sharing a middle hit, *not* the angle to the beam axis. For hits $\mathbf{h}_{k-1}, \mathbf{h}_k, \mathbf{h}_{k+1}$:

$$\alpha_k = \arccos\left(\frac{(\mathbf{h}_k - \mathbf{h}_{k-1}) \cdot (\mathbf{h}_{k+1} - \mathbf{h}_k)}{\|\mathbf{h}_k - \mathbf{h}_{k-1}\| \|\mathbf{h}_{k+1} - \mathbf{h}_k\|}\right). \quad (4)$$

A pair is labelled **true** only when all three hits lie on the same truth track; otherwise it is **false**.

3 Pipeline Summary

3.1 Caching and Persistence

The notebook uses persistent storage under `characterisation_outputs/` to avoid recomputing expensive studies:

- clean event pipeline,
- 10-event batch,
- 100×50 angle distributions,
- parametric scan outputs,
- acceptance histogram study.

A `FORCE_RECOMPUTE` switch controls regeneration.

3.2 Clean Event Reconstruction

A clean event with 3 tracks and 5 modules is generated, solved with `SimpleHamiltonianFast`, then reconstructed with threshold

$$\text{baseline} = \frac{\delta}{\delta + \gamma}, \quad \text{threshold} = \frac{1 + \text{baseline}}{2}, \quad (5)$$

using $\gamma = 1.5$ and $\delta = 1.0$.

Validation is performed with `EventValidator` using purity cut `purity_min=0.7`, reporting event metrics and per-track labels (primary/ghost/clone).

3.3 Multi-Event Studies

10-event varying occupancy. Track counts are scanned over $[2, 3, 4, 5, 6, 3, 4, 5, 2, 4]$, with per-event summaries and overlaid truth/reco plots.

100 events $\times$ 50 tracks. Large-sample true/false segment-angle distributions are accumulated and saved as `angle_distributions_100x50.npz`.

Segment-level scan vs event size. Track multiplicity is scanned from 10 to 100 (step 10), with 5 repeats per point, for:

- scattering multipliers: $1\times, 2\times, 4\times$,
- angular settings: $\phi_{\max} = \theta_{\max} \in \{0.2, 0.1, 0.04\}$ rad.

For each configuration, the study records:

- segment efficiency,
- segment false rate,
- counts of true/false segment pairs,
- counts of accepted true/false pairs with $\theta \leq \varepsilon$.

Acceptance histograms. For each scattering-angle setting, 20 events at 50 tracks/event are generated, with separate true/false histograms and normalised overlays.

4 Metrics

For each event or scan point, the notebook computes:

$$\text{Segment efficiency} = \frac{N_{\text{true, accepted}}}{N_{\text{true pairs}}}, \quad (6)$$

$$\text{Segment false rate} = \frac{N_{\text{false, accepted}}}{N_{\text{all accepted}}}, \quad (7)$$

where acceptance is defined by $\theta \leq \varepsilon$.

Scan summaries report means and standard errors across repeats.

5 Figures and Outputs

The notebook produces the following classes of visual outputs:

- 3D truth event display for a clean event.
- Truth-vs-reconstructed event overlays.
- Per-event and aggregated true/false segment-angle histograms.
- 2×2 metric panels versus event size for each angular setting.
- 3-column acceptance histogram panels (true/false/normalised overlay) for each scattering setting.

If the PNG exports already exist, include them directly. Two known examples in this folder are shown below.

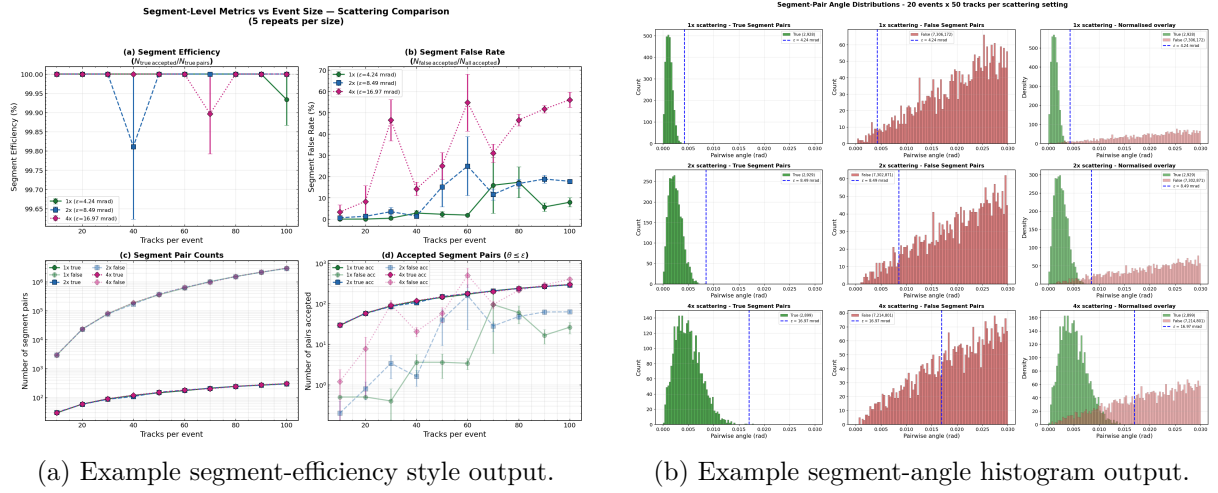


Figure 1: Representative figures exported in the study directory.

6 Reproducibility Notes

- Core randomness comes from event generation; use fixed seeds if strict reproducibility is required.
- Cached outputs are reused unless `FORCE_RECOMPUTE=True`.
- The parametric scans are computationally expensive and are designed to be checkpointed via persistence.

7 How to Extend

Recommended next extensions for this study:

- Add uncertainty bands from bootstrap resampling in addition to standard error.
- Export all figure panels with deterministic filenames and add automatic LaTeX inclusion.
- Compare threshold strategies beyond fixed-sigma ϵ (for example quantile-based acceptance).

8 Conclusion

The notebook implements a complete characterisation pipeline for toy VELO segment-pair selection and reconstruction behaviour. Its main result is a consistent framework to assess how acceptance thresholding, occupancy, angular phase-space, and scattering jointly control efficiency–false-rate trade-offs.